XIAOFENG GUO

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EDUCATION

Carnegie Mellon University — Pittsburgh, U.S.
PhD program in Robotics — Aug. 2020 – present

- GPA: 3.83/4.00
- Advised by Prof. Guanya Shi and Prof. Sebastian Scherer
- Anticipated graduation date: Jan. 2026

Tsinghua University — Beijing, China
Bachelor of Engineering in Mechanical Engineering — Aug. 2016 – Jun. 2020

- GPA: 3.56/4.00
- Visiting summer research intern at Georgia Institute of Technology — Jul. 2019 – Oct. 2019

INDUSTRIAL RESEARCH EXPERIENCE

Amazon FAR (Frontier AI & Robotics) — San Francisco, U.S.
Applied Scientist Intern — May. 2025 – present

- Advised by Prof. Pieter Abbeel and Dr. Rocky Duan
- Research Topics: humanoid whole-body control, dexterous bimanual manipulation

SELECTED PUBLICATIONS AND PATENTS

1. Gupta, H., **Guo, X.**, Ha, H., Pan, C., Cao, M., Lee, D., ... & Shi, G. (2025). UMI-on-Air: Embodiment-Aware Guidance for Embodiment-Agnostic Visuomotor Policies. arXiv preprint arXiv:2510.02614.
2. He, G.*, **Guo, X.***, Tang, L., Zhang, Y., Mousaei, M., Xu, J., Geng, J., Scherer, S. & Shi, G. (2025). Flying Hand: End-effector-centric framework for versatile aerial manipulation teleoperation and policy learning. In *2025 Robotics: Science and Systems (RSS)*. IEEE.
3. **Guo, X.,** He, G., Xu, J., Mousaei, M., Geng, J., Scherer, S., & Shi, G. (2024). Flying Calligrapher: Contact-Aware Motion and Force Planning and Control for Aerial Manipulation. *IEEE Robotics and Automation Letters (RA-L)*.
4. **Guo, X.,** He, G., Mousaei, M., Geng, J., Shi, G., & Scherer, S. (2024, May). Aerial interaction with tactile sensing. In *2024 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 1576-1582). IEEE.
5. **Guo, X.**, Huang, H. J., & Yuan, W. (2023, October). Estimating Properties of Solid Particles Inside Container Using Touch Sensing. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 8985-8992). IEEE.
6. Huang, H. J., **Guo, X.**, & Yuan, W. (2022). Understanding Dynamic Tactile Sensing for Liquid Property Estimation. In *2022 Robotics: Science and Systems (RSS)*. IEEE.
7. Deng, Y., Guo, D., **Guo, X.**, Zhang, N., Liu, H., & Sun, F. (2020). MQA: Answering the question via robotic manipulation. In *2020 Robotics: Science and Systems (RSS)*. IEEE.
8. **Guo, X.**, Blaise, B., Molnar, J., Coholich, J., Padte, S., Zhao, Y., & Hammond, F. L. (2020, May). Soft foot sensor design and terrain classification for dynamic legged locomotion. In *2020 3rd IEEE International Conference on Soft Robotics (RoboSoft)* (pp. 550-557). IEEE.、
9. Deng, Y.*, **Guo, X.***, Wei, Y.*, Lu, K.*, Fang, B., Guo, D., Sun, F., & Liu, H. (2019). Robot Grasping in Cluttered Environment with Active Exploration. In *2019 International Conference on Intelligent Robots and Systems (IROS)*. IEEE.
10. A Chinese patent which has been published and is to be authorized: Sucker-gripper composite grabbing device, Applicant: Tsinghua University, Inventor: Bin Fang, Huaping Liu, **Xiaofeng Guo**, Yuhong Deng, Kai Lu, Yixuan Wei, Public number: 109465840A

RESEARCH EXPERIENCE

UMI on Air: Embodiment-Aware Guidance for Embodiment-Agnostic Visuomotor Policies Feb. 2025 – Sep. 2025

PhD student *CMU* *Advisor: Prof. Guanya Shi and Prof. Sebastian Scherer*

- We proposed Embodiment-Aware Diffusion Policy (EADP), which couples a high-level UMI policy with a low-level embodiment-specific controller at inference time. By integrating gradient feedback from the controller's tracking cost into the diffusion sampling process, our method steers trajectory generation towards dynamically feasible modes tailored to the deployment embodiment. We validated our approach on multiple long-horizon and high-precision real-world aerial manipulation tasks,

Flying Hand: End-Effector-Centric Framework for Versatile Aerial Manipulation Teleoperation and Policy Learning May 2024 – Feb. 2025

PhD student *CMU* *Advisor: Prof. Guanya Shi and Prof. Sebastian Scherer*

- We proposed a unified aerial manipulation framework with an end-effector-centric interface, consisting of a fully-actuated hexarotor with a 4-DoF robotic arm, a whole-body model predictive controller, an intuitive ee-centric aerial teleoperation for versatile tasks and an imitation-learning-based policy for versatile aerial manipulation. Real-world experiments show that the proposed framework significantly improves ee tracking accuracy, and can handle multiple aerial teleoperation and imitation learning tasks, including aerial writing, peg-in-hole, pick and place, light bulb changing, etc.

Flying Calligrapher: Contact-Aware Motion and Force Planning and Control for Aerial Manipulation Sep. 2023 – May 2024

PhD student *CMU* *Advisor: Prof. Guanya Shi and Prof. Sebastian Scherer*

- We proposed a contact-aware trajectory planning algorithm to generate dynamically feasible contact force and motion trajectory in the contact plane. We developed a contact-aware hybrid motion-force control algorithm to enable the UAM to track the continuous time varying contact force and motion reference simultaneously, while compensating for friction force. We developed a system and pipeline for a novel aerial calligraphy task.

Aerial Manipulation with Tactile Sensing Oct. 2022 – Sep. 2023

PhD student *CMU* *Advisor: Prof. Guanya Shi and Prof. Sebastian Scherer*

- We developed a new aerial manipulator system which integrated tactile sensor with a fully-actuated UAV. We proposed a pipeline that leverages tactile feedback for real-time force tracking using a hybrid motion-force controller and a method to utilize a vision-based tactile sensor for wall texture detection during aerial interaction.

Estimating Properties of Solid Particles using Touch Sensing Aug. 2020 – Oct. 2022

PhD student *Carnegie Mellon University* *Advisor: Prof. Wenzhen Yuan*

- We designed and fabricated a high-speed GelSight, a new tactile sensor with both high temporal resolution (815 Hz) and high spatial resolution (640×480). We designed a sequence of actions and designed feature extraction to estimate multiple properties of solid particles, including mass, volume, particle shape, and particle size.

Understanding Dynamic Tactile Sensing for Liquid Property Estimation Aug. 2021 – Jan. 2022

PhD student *Carnegie Mellon University* *Advisor: Prof. Wenzhen Yuan*

- We proposed a simple physics-inspired model to explain the liquid oscillation after perturbation and sense that using tactile sensing. We extracted the decay rate and oscillation frequency of the tactile signals to estimate the liquid volume and viscosity.

MQA: Answering the Question via Robotic Manipulation Oct. 2019 – Aug. 2020
Research Assistant *Tsinghua University* *Advisor: Prof. Huaping Liu*

- We formulated a novel Manipulation Question Answering (MQA) task. We build a novel MQA dataset including a variety of object models, bin scenarios and question-answer pairs, and established a corresponding benchmark. We designed a deep Q-Network for the robot to effectively generate manipulations actions to complete the MQA task.

Foot Sensor Design and Terrain Classification for Dynamic Locomotion Jul. 2019 – Oct. 2019
Research Intern *Georgia Institute of Technology* *Advisor: Prof. Ye Zhao*

- We designed and fabricated a soft contact pad for a legged robot with multiple types of sensors embedded in for acquiring rich sensing information of terrains, using the tactile sensor, acoustic sensor, capacitive sensor, and accelerometers. We performed feature extraction on the multi-modal signals and developed a terrain classification algorithm, which had a high classification accuracy of 96.7%.

Composite Grasping System for Cluttered Environment Oct. 2017 – Dec. 2019
Research Assistant *Tsinghua University* *Advisor: Prof. Huaping Liu*

- We designed and fabricated three kinds of composite robotic hands that combined the suction cup and different grippers for efficient grasping. We designed an object recognition algorithm based on tactile sensing and vision. We developed an efficient grasping strategy and trained an active exploration model which can push down the stacked objects to make the cluttered environment sparse until there is an object predicted to be easily grasped.

SELECTED AWARDS AND HONORS

- 1st Prize at Tsinghua Academic and Science Competition Apr. 2019
- School Scholarship of Tsinghua University: Academic Performance Dec. 2018
- 1st Place at the 20th National Robot and Artificial Intelligence Competition Oct. 2018
- 3rd Place at the 13th ASME Student Mechanism and Robot Design Competition Aug. 2018
- School Scholarship of Tsinghua University: Technological Innovation Dec. 2017

ADDITIONAL INFORMATION

Programming Skills and Software: Python, C++, C, MATLAB, Julia, PyTorch, Isaac Lab, Isaac Gym, MuJoCo, Gazebo, Pybullet, Taichi, ROS/ROS2, SolidWorks, AutoCAD
Computing platforms: Nvidia Jetson, Arduino, MSP430, STM32
Experimental Skills: 3D Print, Laser Cutter, Lathe, Band Saw, Drill, Soldering Station
Languages: Mandarin (native), English (fluent)